

Ising Model Simulation

Objective:

Simulate a 2D Ising model lattice using Monte Carlo Metropolis algorithm.

Method:

- Lattice size = 20x20
- Temperature $T = 2.0$
- Monte Carlo steps = 1000
- Spins flip according to Metropolis criterion:
 - Flip accepted if energy decreases
 - Otherwise accepted with probability $\exp(-dE/kT)$

Result:

- Final lattice configuration shown below.

